On the convergence of mSGD and AdaGrad for stochastic optimization

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Algorithms:

SGD: $\theta_{n+1} = \theta_n - \epsilon_n \nabla_{\theta_n} g(\theta_n, \xi_n)$

mSGD: $v_n = \alpha v_{n-1} + \epsilon_n \nabla_{\theta_n} g(\theta_n, \xi_n),\ \theta_{n+1} = \theta_n - v_n$

AdaGrad: $\theta_{n+1} - \theta_n - \dfrac{}{\sqrt{\sum_{t=1}^{n} \|\nabla_{\theta_n} g(\theta_n, \xi_n)\|^2}} \nabla_{\theta_n} g(\theta_n, \xi_n)$

Contributions:

- We prove that the iterates of mSGD are asymptotically convergent to a connected set of stationary points for possibly non-convex loss function almost surely (i.e., with probability one), which is more general than existing works on subsequence convergence.
- We quantify the convergence rate of mSGD for the loss functions. Through this convergence rate we can get a theoretical explanation of why mSGD can be seen as an acceleration of SGD. Moreover, we provide the convergence rate of mean-square gradients and connect it to the convergence of time averages.
- We prove the iterates of AdaGrad are asymptotically convergent to a connected set of stationary points almost surely for possible non-convex loss functions. The convergence result for the AdaGrad extends the subsequence convergence in the literature.

Main assumptions:

Assumption 1 *Noise sequence $\{\xi_n\}$ are mutually independent and independent of θ_1 and v_0, such that $g(x) = \mathbb{E}_{\xi_n}\left(g(x, \xi_n)\right)$ for any $x \in \mathbb{R}^N$.*

Assumption 2 ***(Loss function assumption)*** *Loss function $g(\theta)$ satisfies the following conditions:*

1) $g(\theta)$ is a non-negative and continuously differentiable function.

2) The set of stationary points of $\|\nabla_\theta g(\theta)\|$ is not an empty set, that is

$$J := \{\theta \mid \|\nabla_\theta g(\theta)\| = 0\} \neq \emptyset.$$

3) $\nabla_\theta g(\theta)$ satisfies the Lipschitz condition, i.e., there is a scalar $c > 0$, such that for any $x, y \in \mathbb{R}^N$

$$\left\|\nabla_x g(x) - \nabla_y g(y)\right\| \leq c\|x - y\|.$$

4) There is a scalar $M > 0$ such that for any $\theta \in \mathbb{R}^N$ and positive integer n,

$$\mathbb{E}_{\xi_n}\left(\left\|\nabla_\theta g(\theta) - \nabla_\theta g(\theta, \xi_n)\right\|^2\right) \leq M\left(1 + g(\theta)\right). \tag{6}$$

Assumption 3 *Momentum coefficient $\alpha \in [0, 1)$ and the sequence of step size ε_n is positive, monotonically decreasing to zero, such that $\sum_{n=1}^{+\infty} \varepsilon_n = +\infty$ and $\sum_{n=1}^{+\infty} \varepsilon_n^2 < +\infty$.*

Assumption 4 ***(Loss function assumption)*** *Loss function $g(\theta)$ satisfies the following conditions:*

1) $g(\theta)$ is a non-negative and continuously differentiable function. The set of its stationary points $J = \{\theta \mid \|\nabla_\theta g(\theta)\| = 0\}$ is a bounded set which has only finite connected components $J_1, \ldots, J_n$. In addition, there is $\varepsilon' > 0$, such that for any $i \in \{1, 2, \ldots, n\}$ and $0 < d(\theta, J_i) < \varepsilon'$, it holds that $\left|g(\theta) - g_i\right| \neq 0$, where $g_i = \{g(\theta) \mid \theta \in J_i\}$ is a constant.

2) For any $i \in \{1, 2, \ldots, n\}$, it holds that

$$\liminf_{d(\theta, J_i) \to 0} \frac{\|\nabla_\theta g(\theta)\|^2}{g(\theta) - g_i} \geq s > 0.$$

Assumption 5 *Loss function $g(\theta)$ in equation 5 satisfies the following conditions:*

1) $g(\theta)$ is a non-negative and continuously differentiable function. The set of its stationary points $J = \{\theta \mid \|\nabla_\theta g(\theta)\| = 0\}$ is a bounded set which has only finite connected components $J_1, \ldots, J_n$. In addition, there is $\varepsilon_1 > 0$, such that for any i and $0 < d(\theta, J_i) < \varepsilon_1$, it holds that $\left|g(\theta) - g_i\right| \neq 0$, where $g_i = \{g(\theta) \mid \theta \in J_i\}$ is a constant.

2) The gradient $\nabla_\theta g(\theta)$ satisfies the Lipschitz condition, i.e., for any $x, y \in \mathbb{R}^N$,

$$\left\|\nabla_x g(x) - \nabla_y g(y)\right\| \leq c\|x - y\|.$$

3) There are two constants $M' > 0$ and $a > 0$ such that for any $\theta \in \mathbb{R}^N$ and $n \in \mathbb{N}_+$,

$$E_{\xi_n}\left(\left\|\nabla_\theta g(\theta, \xi_n)\right\|^2\right) \leq M'\left\|\nabla_\theta g(\theta)\right\|^2 + a. \tag{11}$$

Main results:

Theorem 1 *Consider the mSGD in equation 3. If Assumptions 1–3 hold, then for $\forall \theta_1 \in \mathbb{R}^N$ and $\forall v_0 \in \mathbb{R}^N$, there exists a connected set $J^* \subseteq J$ such that the iterate θ_n is convergent to the set J^* almost surely, i.e.,*

$$\lim_{n \to \infty} d(\theta_n, J^*) = 0, \qquad a.s.$$

where $d(x, J^*) = \inf_y\{\|x - y\|, y \in J^*\}$ denotes the distance between point x and set J^*.

Theorem 2 *Consider the mSGD in equation 3 with the noise following a uniform sampling distribution. If Assumptions 1-4 hold, for $\forall v_0 \in \mathbb{R}^N$ and $\forall \theta_1 \in \mathbb{R}^N$, it holds that*

$$\mathbb{E}\left(\|\nabla_{\theta_n} g(\theta_n)\|^2\right) = O\left(e^{-\sum_{i=1}^{n} \frac{s\varepsilon_i}{p(1-\alpha)^2}}\right), \tag{10}$$

where $p = \exp\left\{\sum_{k=1}^{\infty} M\varepsilon_k^2\right\}$ and M is defined in condition 4) of Assumption 2.

In Theorem 2, let $\alpha = 0$, then we can obtain the convergence rate of SGD,

$$\mathbb{E}\left(\|\nabla_{\theta_n} g(\theta_n)\|^2\right) = O\left(e^{-\sum_{i=1}^{n} \frac{s}{p}\varepsilon_i}\right).$$

Theorem 3 *Consider the AdaGrad in equation 5. If Assumptions 1 and 5 hold, then for $\forall \theta_1 \in \mathbb{R}^N$ and $S_0 = 0$, there exists a connected component of the set $J^* \subseteq J$, such that the estimate θ_n is convergent to the set J^* almost surely, i.e.,*

$$\lim_{n \to \infty} d(\theta_n, J^*) = 0. \qquad a.s.$$

Proof outline for Theorems 1-2:

The proof is in light of the Lyapunov method. We aim to prove that $g(\theta_n)$ is convergent a.s., and then to prove $\nabla_{\theta_n} g(\theta_n) \to 0$ *a.s.* With these two results, we are able to get $\theta_n \to J^*$ *a.s.* In the following, we provide the proof outline to show how to obtain the provided results of mSGD.

Step 1: We prove mSGD is a stable algorithm, i.e., $\mathbb{E}(g(\theta_n)) < K < +\infty, \forall\, n$, in Lemma 1. The idea is to prove that a weighted sum of the loss function value, i.e.,

$$U_n := \sum_{t=1}^{n} \left(1/(2-\alpha)\right)^{n-t} \mathbb{E}\left(1 + g(\theta_{t+1})\right)$$

is bounded through a recursion formula (a rough form)

$$U_n - U_{n-1} \leq A\alpha^n + \underbrace{B\sum_{t=1}^{n-1} \left(1/(2-\alpha)\right)^{n-t} \varepsilon_t^2\, \mathbb{E}\left\|\nabla_{\theta_t} g(\theta_t, \xi_t)\right\|^2}_{I} \ (A,\ B \textit{ are two constants}).$$

Then we apply Assumption 2 4) to I and then obtain

$$U_n - U_{n-1} \leq A\alpha^n + \underbrace{B\sum_{t=1}^{n-1} \left(1/(2-\alpha)\right)^{n-t} \varepsilon_t^2\, \mathbb{E}\left(1 + g(\theta_t)\right)}_{R} \ (A,\ B \textit{ are two constants}).$$

Combining U_{n-1} and R leads to

$$F_n - F_{n-1} \leq A\alpha^n + B'\left(\frac{1}{2-\alpha}\right)^n \ (A,\ B' \textit{ are two constants}),$$

where

$$F_n := \sum_{t=1}^{n} \left(\frac{1}{2-\alpha}\right)^{n-t} Z(t+1)\, \mathbb{E}\left(e_{t+1}^{(n)}\left(1 + g(\theta_{t+1})\right)\right)$$

and

$$Z(t) = \prod_{k=t}^{+\infty}(1 + M_0\varepsilon_k^2) = (1 + M_0\varepsilon_t^2) \prod_{k=t+1}^{+\infty} (1 + M_0\varepsilon_k^2) = (1 + M_0\varepsilon_t^2)Z(t+1).$$

Thus, we are able to obtain $E(g(\theta_n)) < F_{n-1} \leq A\sum_{t=1}^{+\infty} \alpha^t + B\sum_{t=1}^{+\infty} \left(1/(2-\alpha)\right)^t < +\infty$.

Step 2: From Lemma 1 and the condition $\sum_{t=1}^{+\infty} \varepsilon_n^2 < +\infty$, we are able to prove that $\sum_{t=1}^{n} \|v_t\|^2$ and $\sum \varepsilon_t \|\nabla_{\theta_t} g(\theta_t)\|^2$ are convergent a.s. respectively, as stated in Lemma 7 and Lemma 8.

Step 3: We divide $g(\theta_n)$ into three terms

$$g(\theta_n) = \sum_{t=1}^{n} A(n)\|v_t\|^2 + \sum_{t=1}^{n} B_n\varepsilon_t \|\nabla_{\theta_n} g(\theta_n)\|^2 + \sum_{t=1}^{n} C_n^T\left(\nabla_{\theta_n} g(\theta_n, \xi_n) - \nabla_{\theta_n} g(\theta_n)\right).$$

From Lemma 7 and Lemma 8, we are able to prove that $\sum_{t=1}^{n} A(n)\|v_t\|^2 + \sum_{t=1}^{n} B_n\varepsilon_t \|\nabla_{\theta_n} g(\theta_n)\|^2$ is convergent a.s.. From the convergence theorem for martingale-difference sum (Lemma 5), we prove that $\sum_{t=1}^{n} C_n^T\left(\nabla_{\theta_n} g(\theta_n, \xi_n) - \nabla_{\theta_n} g(\theta_n)\right)$ is convergent a.s. Then we prove $g(\theta_n)$ is convergent a.s.

Step 4: By Lemma 9 and the convergence of $g(\theta_n)$ in Step 3, we get $\theta_n \to J^*$ a.s..

Step 5: After the proof of the convergence of mSGD, we analyze the iterates of F_n. Then under a new assumption $\liminf_{d(\theta, J_i) \to 0} \|\nabla_\theta g(\theta)\|^2 / \left(g(\theta) - g_i\right) \geq s > 0$, we are able to obtain the convergent rate of mSGD.